

Retrieval and Answer-Key Integrity Audit of a CompTIA Security+ Study Note in a Personal Obsidian Vault

A Single-Participant Session Analysis

Anonymous Author

Session record: “Locate CompTIA+ Book in Obsidian Vault”

25 September 2026

Abstract

This paper reports a small-scale, single-participant audit carried out on one study note held in a personal Obsidian vault, together with a verification of a third-party answer key supplied for that material. The work proceeded in six stages. Stage A performed a case-insensitive recursive retrieval against the vault, which returned a single matching title: a Security+ textbook by Mark Ciampa, present both as a source PDF and as a directory of fifteen extracted per-module notes. Stage B selected Module 7, “Identity and Access Management” (SY0-701), read it in full at 471 lines, and audited it for internal coherence across its three stated objectives — authentication credential types, authentication best practices, and access control schemes — surfacing two defects: an internal contradiction between a statement of industry password-expiry practice and an explicit recommendation against expiry, and a malformed review question presenting two textually identical answer options. Stage C had the participant answer twenty-five examination items under a drill protocol requiring that every answer be scored and accompanied by a line-level citation into the source note; all twenty-five were resolved against the note, and a third defect was identified, an item demanding two selections for which only one was supportable from the source. Stages D–F then examined a purported answer key. In roughly three quarters of its entries the printed option letter contradicted the printed explanatory rationale, and several entries were topically unrelated to any supplied item; re-mapping the key by subject matter rather than by position established coverage of only twenty-one items, leaving four unaddressed. The analysis nonetheless exposed a pattern in which certain printed letters were displaced while the accompanying rationales were correct, motivating the hypothesis that the rationales were authoritative and the letters had been shuffled independently of them. The participant then reissued the item set with all options reordered within each question; under the new ordering the same key became internally consistent, confirming the hypothesis. The reissued set also substituted five new items for four the key had never covered. Rescoring the original responses against the now-aligned key produced thirteen confirmed correct responses, three confirmed incorrect responses, four partial or disputed responses, and five newly covered items. The three confirmed errors are analysed individually, and the disputed items are preserved as open findings rather than resolved. The methodological conclusions are that an answer key’s option letters are not self-validating, that a rationale must be checked for consistency with both the item and the letter before use, that positional alignment must be verified before scoring, and that fidelity to a single source can produce confident errors whenever that source is silent on the distinction actually being tested.

Contents

1 Introduction

2

1.1	Motivation	2
1.2	Research questions	3
1.3	Contributions and non-contributions	3
2	Material Under Audit	3
2.1	Retrieval outcome	3
2.2	The selected note	4
3	Methods	4
3.1	Stage A — Retrieval and scoping	4
3.2	Stage B — Full read and coherence audit	4
3.3	Stage C — Item response under a citation protocol	4
3.4	Stage D — Answer-key integrity analysis	4
3.5	Stage E — Order-randomised reissue	5
3.6	Stage F — Rescoring	5
3.7	Reporting conventions	5
4	Results	5
4.1	Note defects surfaced by the full read	5
4.2	Item response coverage	6
4.3	Answer-key integrity	6
4.4	Confirmation of the shuffle hypothesis by option reordering	6
4.5	Rescoring outcome	6
4.6	The three confirmed errors	7
4.7	Partial and disputed items	7
4.8	Newly covered items	8
5	Discussion	9
5.1	Option letters are not self-validating	9
5.2	Rationale–letter–item consistency is a precondition for scoring	9
5.3	Positional alignment must be verified before scoring	9
5.4	Single-source fidelity can produce confident errors	9
5.5	Threats to validity	9
5.6	Open findings	10
6	Conclusion	10

1 Introduction

1.1 Motivation

Personal knowledge management systems increasingly serve as the substrate for certification exam preparation. When a certification guide is imported into a vault and decomposed into per-module notes, two properties become immediately consequential. The first is *structural*: the learner must be able to find the intended artifact, and must be able to identify which fragment

of a large imported corpus corresponds to the objective currently under study. The second is *epistemic*: material presented in the form of an assessment, together with any key supplied alongside it, must be internally consistent, otherwise the learner rehearses against error rather than against the underlying subject.

The second property is the more fragile of the two. A note can be located reliably and still teach an inconsistency. An answer key can be handed over in good faith and still be internally contradictory, and the contradiction is not visible to a learner who reads only the letter. The failure mode is silent: the learner records a verdict, carries it forward as settled, and does not discover the error until an examination disagrees.

1.2 Research questions

This session record is organised around four questions, each answerable from the audit trail alone.

1. **Retrieval and scoping.** Can the intended study artifact be located unambiguously within a personal vault, and can the note corresponding to a named module objective be identified and read to completion?
2. **Note integrity.** What classes of defect are detectable in a study note by a full read against a single declared source of truth, and which concrete defects were found in the note audited here?
3. **Key integrity.** When a purported answer key is checked against the item set and the source note, in what ways does it fail, and can the nature of the failure be characterised rather than merely reported?
4. **Scoring validity.** What does rescoring a recorded response set against a key whose internal structure has been repaired reveal, and which responses are confirmed, refuted, or left genuinely open?

1.3 Contributions and non-contributions

The contribution is a methodological case record: a single artefact audited to exhaustion, with the defect classes, the key-corruption pattern, and the rescoring outcome documented descriptively. The non-contributions are equally deliberate. No pass rates, score statistics, effect sizes, or reliability estimates are computed, because a single participant working a single twenty-five-item set does not license them. No external literature is cited, because none was consulted. Where the session record characterises an item only partially, that partiality is carried forward as an open finding rather than resolved by inference.

2 Material Under Audit

2.1 Retrieval outcome

A case-insensitive recursive search was executed against the vault. Exactly one title matched the CompTIA study-book query: a Security+ textbook by Mark Ciampa [1]. The artifact exists in two representations — a source PDF, and a directory of fifteen extracted per-module notes. This dual representation is the structural point of interest: the learner must know which representation to consult, and the per-module decomposition is what makes a single objective addressable at all.

2.2 The selected note

Scope was narrowed by the participant to Module 7, “Identity and Access Management” (SY0-701). The note was read in full at 471 lines. It is organised around three stated objectives:

1. authentication credential types;
2. authentication best practices;
3. access control schemes.

The three-objective structure matters methodologically, because it supplies the partition against which later items were classified. An item about a credential type, an item about a practice recommendation, and an item about an access control scheme are each adjudicable against a distinct region of the note, which is what makes a line-level citation a meaningful check rather than a gesture.

3 Methods

The audit was executed as six sequential stages. Each stage consumed the output of its predecessor, so the design is strictly chained: a failure at any stage would have invalidated the stages downstream of it.

3.1 Stage A — Retrieval and scoping

A case-insensitive recursive search was run over the vault for the CompTIA study book. The single returned title was then decomposed into its available representations, and the module directory enumerated. The participant selected Module 7 by name, fixing the scope of all subsequent work.

3.2 Stage B — Full read and coherence audit

The selected note was read to completion rather than sampled. The audit sought contradictions internal to the note — statements that cannot both be true, or question forms that cannot be answered as written — and did not import expectations from outside the note. The point of this stage is that the note is its own reference standard: a defect is a defect regardless of what external sources say.

3.3 Stage C — Item response under a citation protocol

Twenty-five examination items were supplied and answered. The drill protocol imposed two obligations on every response: the answer was scored, and it was accompanied by a line-level citation into the source note. The citation obligation converts an answer from an assertion into a checkable claim — the recorded response is right or wrong relative to the note, not relative to the learner’s recall. All twenty-five items were resolved against the note under this protocol.

3.4 Stage D — Answer-key integrity analysis

A purported answer key was then supplied. The analysis proceeded in two passes. The *internal* pass compared, for each key entry, the printed option letter against the printed explanatory rationale, looking for entries in which the letter selected an option the rationale itself did not

support. The *coverage* pass then asked a different question — not *is entry k consistent* but *which supplied item does entry k address* — and re-mapped entries to items by subject matter rather than by position. The two passes answer different defects, and only the second can detect a key that has drifted relative to the item set.

3.5 Stage E — Order-randomised reissue

The first key analysis produced an apparent regularity: in a number of entries the letter appeared displaced while the rationale appeared correct. This motivated the hypothesis that the rationales were authoritative and that the letters had been shuffled independently of them. The hypothesis makes a sharp, falsifiable prediction. If it is correct, then the same key, presented against an item set whose options have been reordered within each question, should become internally consistent — the same rationales will now agree with the same letters, because the shuffling that displaced them has been absorbed by the reordering.

The participant accordingly reissued the item set with all options within each question re-ordered. The same key was then re-examined.

3.6 Stage F — Rescoring

With the key’s internal structure established, the originally recorded responses were rescored against it. Each response received one of four dispositions, and the disposition was assigned from the source note and the aligned key together, never from the printed letter alone.

3.7 Reporting conventions

Three conventions are used throughout and are stated here so the results can be read correctly. First, counts reported as *approximately* or *roughly* are reproduced as such from the session record and have not been sharpened. Second, a *disputed* item is one where the note and the aligned key do not jointly determine a single defensible answer; it is reported as disputed, not resolved. Third, an item *newly covered* by the reissued set is not scored, because no recorded response exists for it; it is reported as a coverage gap that the reissue closed.

4 Results

4.1 Note defects surfaced by the full read

Two defects were found in the note before any item was answered.

Defect N-1: password-expiry contradiction. The note contains an internal contradiction between a statement of industry password-expiry practice and an explicit recommendation against expiry. A note that both reports prevailing practice and counsels against it, without reconciling the two, cannot be used to adjudicate a question about whether passwords should expire. This defect later proved consequential: it is the same distinction on which one of the rescored items turned out to be disputed (Section 4.7).

Defect N-2: malformed review question. One review question presents two textually identical answer options. Such an item is unanswerable as written; no selection distinguishes the options, and a recorded response to it carries no information.

4.2 Item response coverage

All twenty-five supplied items were resolved against the note under the citation protocol, and the act of resolving them surfaced a third defect.

Defect N-3: unsupportable dual selection. One item demanded two selections, but only one of them was supportable from the source note. The item’s form over-promises relative to its content.

The three note defects therefore belong to two distinct classes. N-1 is a *contradiction*: two statements in the note cannot both be true, or cannot both be acted on. N-2 and N-3 are *malformations*: the question form is defective — respectively, degenerate options and an unsatisfiable selection cardinality — independent of the truth of any proposition. Only N-1 requires substantive judgement; N-2 and N-3 are detectable by inspection.

4.3 Answer-key integrity

The purported key failed on two independent axes.

Internal consistency. In roughly three quarters of the key’s entries, the printed option letter contradicted the printed explanatory rationale — the letter selected an option that the rationale did not support. A key in this state is worse than no key: a learner who trusts the letter rehearses the opposite of what the explanation teaches, and a learner who trusts the explanation discards the letter.

Coverage. Re-mapping entries to items by subject matter rather than by position established coverage of only twenty-one of the twenty-five items, leaving four unaddressed. Several entries were topically unrelated to any supplied item, meaning the key contained material for which no question had been posed — and, by the same token, questions for which the key provided no answer. The positional mapping the key’s own numbering implied was thus not merely misaligned but incomplete.

4.4 Confirmation of the shuffle hypothesis by option reordering

The reissued item set, with all options within each question reordered, produced the predicted result: the key became internally consistent. The same rationales that had contradicted their letters now agreed with them.

This is a clean confirmation rather than an approximate one, and it is worth being precise about what was established. The hypothesis claimed two things jointly: that the rationales were reliable, and that the letters had been shuffled relative to them. Internal consistency after reordering is evidence for the first claim, because a corrupted rationale would remain corrupted under any reordering. It is evidence for the second, because the only mechanism that can make a fixed set of rationales agree with a fixed set of letters under one ordering and disagree under another is a relative displacement between them. What it does *not* establish is the mechanism of the shuffle — how it occurred, whether by design, transcription error, or generation error — and the session record does not determine this. The finding is a structural diagnosis, not an attribution.

4.5 Rescoring outcome

Rescoring the original twenty-five recorded responses against the now-aligned key produced the following distribution.

Table 1: Disposition of the originally recorded responses after key realignment. Counts are as reported in the session record; $13 + 3 + 4 + 5 = 25$ accounts for the full original item set.

Disposition	Count	Basis
Confirmed correct	13	Response and aligned key agree; the note supports both.
Confirmed incorrect	3	Response conflicts with the note; the defect is identified.
Partial / disputed	4	Note and aligned key do not jointly determine one answer.
Newly covered	5	Reissued items with no recorded response; not a score.
Total	25	

The arithmetic closes: the three confirmed errors, the four disputed items, and the five newly covered items together account for the twelve original items that were not straightforwardly correct, leaving thirteen confirmed correct. The distribution should be read as a statement about the interaction between a defective key and a faithful response set — not as a performance statistic, and not as a basis for any estimate of the learner’s knowledge.

4.6 The three confirmed errors

Each confirmed error is diagnostic, and each is diagnostic in a different way.

1. **E-1: the rainbow-table rejection.** A response selecting the comparison option that consults a database of precomputed hashes was recorded as incorrect, on the grounds that the source note describes plaintext word candidates and never mentions rainbow tables. The rejection was faithful to the note and wrong on the merits. This is the signature error of single-source adjudication: the note enumerates one attack (dictionary-style matching against plaintext word candidates) and is silent on a closely related one, and the adjudicator treated silence as absence. A learner reasoning this way will correctly reject an option the examination expects to be selected.
2. **E-2: the pipeline-step substitution.** A response to a question about the goal of mask usage gave the mask-creation step that precedes it in the pipeline. The recorded response identified a real stage of the correct process but answered a different question — what the mask is for, rather than what makes it. This error is not a knowledge gap in the pipeline; it is a misreading of the interrogative, and it would survive re-study of the same note.
3. **E-3: the contestable security-key letter.** A response selected a security-key option on the grounds of the source note, against the letter printed in the key. The key’s rationale for that item was judged correct; its letter was judged contestable. This is the residual case left after realignment — the letter remains unreliable even where the rationale does not, and adjudication must fall back to the note.

4.7 Partial and disputed items

Four items fell into the partial or disputed class. The session record characterises two of them individually; the remaining two are reported at the class level only.

P-1: password expiration. The key required password expiration. The source note explicitly recommends against it and instead requires a minimum password age — that is, it endorses a floor on password lifetime while rejecting a scheduled ceiling. The key’s requirement and

the note’s recommendation are incompatible, and the incompatibility is the same one recorded as Defect N-1. This item is not resolved here: it is reported as a conflict between the supplied assessment instrument and its declared source of truth, and it is left open for adjudication against the source material as a whole rather than against the note alone.

P-2: biometric scanner specialization. The key’s rationale was contradicted by the note’s own scanner-type table. The statement the key selected — concerning how retinal patterns change — is true in general, but it does not appear in the source note. The item therefore presents a three-way tension: the key’s letter, the key’s rationale, and the note’s table do not agree, and the selected statement is defensible on external grounds but unsupportable from the declared source. It is left open.

P-3, P-4: uncharacterised. Two further items were classified as partial or disputed, but the session record does not characterise their content. They are reported here as open findings and are not resolved by inference.

It is worth noting the structural commonality between P-1 and P-2: in both, the point under test is a distinction on which the declared source is either silent (P-2) or self-contradictory (P-1). This is the same condition that produced Error E-1, and it is why the two classes are reported together in the Discussion.

4.8 Newly covered items

The reissued set substituted five new items for four the earlier key had never covered. These are reported as coverage gaps that the reissue closed, and are not scored, because no response was recorded against them.

N-1: suspected multi-factor fatigue attack. An item asking for the appropriate response to a suspected multi-factor fatigue attack.

N-2: digest-comparison safeguard. An item asking to identify the safeguard protecting a stored banking credential at the point of digest comparison — that is, the mechanism that defeats precomputed comparison of stored digests.

N-3: MAC alongside DAC. An item asking how mandatory integrity control coexists with discretionary access control on a modern operating system.

N-4: privileged-access justification and session revocation. An item asking to identify the enterprise mechanism that records justification for privileged access and revokes sessions.

N-5: privilege inheritance by executed programs. An item asking to identify the access control scheme whose characteristic weakness is that executed programs inherit the account holder’s privileges.

Taken together, the five new items span the note’s three declared objectives more evenly than the original set did: N-1 and N-2 concern authentication, N-3 concerns the coexistence of control types, and N-5 concerns the weakness of a named access control scheme. The substitution is best read as a repair of the coverage map rather than as a change of difficulty.

5 Discussion

5.1 Option letters are not self-validating

The central methodological finding is that an answer key’s option letter carries no independent authority. A letter is a position-dependent token: its referent is fixed by the order of the options printed beside it, and it is therefore invalidated by any reordering, any transcription slip, or any partial regeneration of the option list. The key audited here failed in exactly this way, in roughly three quarters of its entries. The practical consequence is that a key must be treated as a set of *rationales* with letters attached, not as a set of letters with commentary attached — and the rationales must be checked for internal consistency with their own letters before either is used.

5.2 Rationale–letter–item consistency is a precondition for scoring

Each of the three key-analysis passes tested something different: letter against rationale (internal consistency), entry against item set (coverage), and key against reordered item set (structural alignment). A key defective in any one of these three ways will produce wrong scores if used directly, and the three failure modes are not detectable by the same inspection. Positional mapping in particular is the most dangerous, because it looks correct — every entry has a plausible-looking letter in a plausible-looking position — while silently substituting answers across items.

5.3 Positional alignment must be verified before scoring

The re-mapping pass established coverage of only twenty-one of twenty-five items. Had the key been scored positionally, as its numbering invites, four items would have received answers belonging to other questions. The lesson generalises beyond this session: before any response set is scored against any key, the correspondence between key entries and items must be established by content and confirmed, not assumed from ordering.

5.4 Single-source fidelity can produce confident errors

Errors E-1, E-3, and disputed items P-1 and P-2 share a single root condition: the adjudicator was working correctly within a frame in which the source is the only permitted authority, and the source is silent or self-contradictory on precisely the distinction under test. Fidelity to a source is therefore not a guarantee of correctness, and the failure is the dangerous kind — the reasoning is transparent, the citation is present, the confidence is high, and the conclusion is wrong. The citation protocol did its job for what the note does say; it could not adjudicate what the note does not say. A protocol that requires a citation should be supplemented by an explicit rule for the uncitable case: where the source is silent, the response is *not established*, and must be escalated rather than scored against a proxy.

5.5 Threats to validity

The findings are single-participant, single-note, single-session, and based on a twenty-five-item set. They describe one artefact and one key. No claim is made that the corruption pattern generalises to other keys or other corpora — though the mechanism identified in Section 4 (rationale–letter displacement) is a structural property that could recur. The exact proportion of letter–rationale contradictions is reported as approximate; the session record does not supply

a precise count, and none has been estimated. No comparison group, no repeated measure, and no external benchmark was available.

5.6 Open findings

Four findings are deliberately left unresolved.

1. The mechanism by which the key's letters became displaced from its rationales is undetermined; the reordering test establishes the displacement, not its cause.
2. Defect N-1 (the password-expiry contradiction) is unrepaired in the note, and the same contradiction drives the open status of P-1. Repairing the note would likely resolve the item; neither has been done here.
3. Defects N-2 and N-3 (degenerate duplicate options; unsupportable dual selection) are unrepaired in the note and remain latent traps for any future learner.
4. Disputed items P-3 and P-4 are uncharacterised in the available record and are reported at the class level only.

6 Conclusion

A single study note in a personal Obsidian vault was retrieved unambiguously, read to completion at 471 lines, and audited against itself. Three defects were found, in two classes: one substantive contradiction, and two question-form malformations. Twenty-five examination items were answered under a protocol requiring a line-level citation for every response, all twenty-five were resolved, and one of the five substitution items introduced in the reissued set later served to test a coverage gap the original key had left open.

The substantive result concerns the answer key. A purportedly supplied key was internally contradictory in roughly three quarters of its entries and addressed only twenty-one of twenty-five items. Content-based re-mapping revealed a pattern — displaced letters attached to correct rationales — which was confirmed by the decisive test of reissuing the item set with all options reordered, under which the same key became internally consistent. Rescoring against the aligned key yielded thirteen confirmed correct responses, three confirmed incorrect, four partial or disputed, and five newly covered.

The transferable conclusions are four. Option letters in an answer key are not self-validating and must be verified against their own rationales. A rationale must be checked for consistency with both the item and the letter before it is used. Positional alignment between key and item set must be verified by content before scoring begins. And fidelity to a single declared source, however scrupulously applied, produces confident errors wherever that source is silent on the distinction being tested — the uncitable case must be recognised as such and escalated, not scored against a proxy. The four open findings recorded in Section 5 are left open deliberately.

References

- [1] M. Ciampa. *CompTIA Security+ Study Resource* (title as recorded in the vault index), Module 7: “Identity and Access Management” (SY0-701). Extracted per-module note held in a personal Obsidian vault; module note read in full at 471 lines.